Oussama El Asri

AI Engineering & Backend Development · Seeking Internship

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SUMMARY

Computer Science student at 1337 (42 Network) with hands-on experience building AI-powered backend services, RAG pipelines, and LLM-driven agents. Strong foundation in systems programming, API design, and containerized infrastructure. Self-driven learner who ships complete, peer-reviewed projects end-to-end — from architecture to deployment.

TECHNICAL SKILLS

Languages	Python, C, C++ (C++98), JavaScript, TypeScript
AI / LLM	RAG pipelines, LangChain, Ollama, AI Agents, vector embeddings, prompt engineering
Backend / APIs	Flask, FastAPI, Gunicorn, Uvicorn, REST API design
DevOps & Infra	Docker, Docker Compose, NGINX, Git, Linux
Systems	HTTP/1.1, TCP/IP, CGI, non-blocking I/O, concurrency (mutex / semaphore)
Concepts	Data structures, AST construction, parsing algorithms, API design

PROJECTS

AI Studio — Unified AI Platform

Personal · Full-stack

Python · Flask · FastAPI · Gunicorn · Uvicorn · React · TypeScript · Docker Compose · Ollama · LangChain

- Designed and built a full-stack AI platform combining two independent backend services into a single React frontend — full end-to-end ownership from architecture to integration.
- RAG Assistant (Flask + Gunicorn): scrapes any user-provided URL, generates vector embeddings using *nomic-embed-text*, and answers natural-language questions grounded in that content — with fallback to general knowledge mode when context is insufficient.
- Crypto Analysis Agent (FastAPI + Uvicorn): takes natural-language prompts, fetches 30-day OHLCV market data, computes technical indicators (EMA, RSI, Bollinger Bands), and returns AI-generated market analysis.
- Orchestrated all services via Docker Compose with a shared Ollama LLM runtime and isolated container networking — zero external API dependencies, fully local inference.
- Intentionally used Flask and FastAPI across the two services to gain hands-on experience with both ecosystems and their integration patterns.

Webserv — HTTP/1.1 Web Server

Team project · 3 members

C++98 · poll() · CGI · NGINX-inspired config

- Owned the HTTP response generation module: full GET, POST (including file upload), and DELETE handling — status codes, response headers, payload construction, and clean error pages.
- Implemented CGI script execution via Python, handling environment variable setup, process forking, chunked request un-chunking, and EOF-terminated output parsing.
- Collaborated across defined team interfaces — request parsing, non-blocking I/O loop, and response — with full HTTP/1.1 spec compliance verified against NGINX.

cub3D — First-Person Raycasting Engine

Team project · 2 members

C · miniLibX · DDA algorithm · XPM textures

- Implemented the full raycasting engine using the DDA (Digital Differential Analyzer) algorithm — computing ray intersections, wall distances, and projected wall heights in real time.
- Rendered directional wall textures (N/S/E/W), configurable floor and ceiling colors, and smooth player movement and camera rotation from scratch using miniLibX.

Other Projects

- **Minishell** (C) — Unix shell with command parsing, piping, redirections, env variable expansion, and built-ins. Full process lifecycle management using fork/exec/wait.
- **Philosophers** (C) — Dining Philosophers problem using POSIX mutexes and semaphores. Deadlock-free synchronization with precise timing constraints.
- **JSON Parser** (Python) — Recursive-descent parser building a full AST without Python's json module, converting to native Python types.
- **Inception** (Docker) — Multi-container production environment: NGINX with TLS termination, WordPress, and MariaDB with persistent volumes.

EDUCATION

1337 School — 42 Network

Morocco · 2023 – Present

Computer Science · Project-Based, Peer-to-Peer Learning

Part of the global 42 Network founded by Xavier Niel. No professors — learning is driven entirely by self-initiative, peer code reviews, and project delivery. Curriculum spans algorithms, systems programming, networking, DevOps, and AI engineering.

LANGUAGES

Arabic — Native   **English** — Intermediate   **French** — Intermediate